

Bhavdeep Arora

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EDUCATION

Toronto Metropolitan University (Formerly Ryerson University)

Toronto, ON

Bachelor of Science (Honours) Computer Science · Software Engineering Concentration

2022–2026

- Relevant Coursework: Data Structures & Algorithms, Software Engineering, Operating Systems, Data Science

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript, HTML/CSS, Go, Rust

Frameworks: React, Node.js, Tailwind, Django, Flask

Developer Tools: Git, Docker, Google Cloud Platform, TravisCI

Data & Cloud: Hadoop, Apache Spark, MySQL, MongoDB, SQLite, AWS, Azure, Firebase

PROJECTS

Thunderhead | Go, Networking, Reverse Proxy, Security | [Github](#)

- Built a **lightweight reverse proxy** in Go that analyzes incoming HTTP requests and assigns an **intent score** (0–100) to detect and mitigate bot traffic without relying on third-party services
- Designed a **behavioral scoring engine** using signals such as request rate, path traversal patterns, header anomalies, and robots.txt violations to classify traffic
- Implemented **graduated response handling** including request forwarding, tarpit delays, and 403 blocking, along with **structured JSON logging** for observability

Deadzone | Python, Random Forest, ERA5/NOAA Datasets, Plotly Dash | [Github](#)

- Engineered an end-to-end **ML pipeline** analyzing Gulf of Mexico hypoxic dead zones using ERA5 and NOAA datasets, training a **Random Forest model** to predict dead zone extent from climate and nutrient runoff features
- Built an **interactive Plotly Dash dashboard** to visualize spatial dead zone trends and model outputs across multi-decade environmental data

Terraseed | Python, Machine Learning, Climate Datasets, Plotly Dash | [Github](#)

- Developed a **global planting window predictor** using ML classification models trained on large-scale gridded climate data to generate region-specific crop recommendations by temperature, precipitation, and frost risk
- Delivered results through an **interactive geospatial dashboard** allowing users to query optimal planting windows by location and crop type

Redis Implementation | C++, TCP/IP, Data Structures, Networking | [Github](#)

- Built a high-performance **in-memory data structure server** in C++ with multiple data types (strings, lists, sets, hashes) and a **TCP networking layer** handling concurrent client connections via the RESP protocol
- **Implemented persistence and optimization features** including RDB snapshots, AOF logging, custom memory allocators, and LRU eviction for efficient cache management

VOLUNTEER EXPERIENCE

Inspire Event 2023 | Speaker Coordinator Team Leader

January 2023 - March 2023

- Led a team of coordinators to source and onboard speakers for a university-scale event, managing outreach and scheduling end-to-end
- Ensured seamless day-of execution by overseeing speaker logistics, timing, and audience engagement across sessions

Sci Exchange at Toronto Metropolitan University | Volunteer & Assistant

September 2023 - April 2024

- Mentored and guided 50+ participants through science exchange exhibits, translating complex topics into accessible explanations
- Collaborated with organizing team to coordinate event flow and participant experience across a multi-hour program